External validation and evidence-gated decision support for a virtual hydrogen refuelling station digital twin

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Working manuscript. The external expert review of SAGA decision support has not been completed. The manuscript is therefore not ready for journal submission, and no placeholder effectiveness result may be interpreted as evidence.

Abstract

This study evaluates a virtual hydrogen refuelling station using separate evidence gates for process physics, consequence software and language-model decision support. A Type-IV tank model yielded validation errors of 3.841 MPa, 4.833 °C and 3.850 state-of-charge percentage points across 12 frozen fills, but did not transfer to the full controller. A public 15-workbook intake contained 58,440 synchronized samples; 8/9 comparable post-access transfer sessions passed descriptive flow–mass closure (18 sessions detected). A 22-case PRESLHY blowdown evaluation passed 50.0% against a 70% threshold; a non-adiabatic revision passed 20/22 consumed cases but only 2/3 eligible holdouts. Independent 90 MPa releases passed 0/3 mass-flow series. HyRAM+ matched its API and an ignited-enclosure model passed 27/27 peak-pressure screens. A 44-case large-scale delayed-ignition rank screen passed pooled correlation but failed all geometry-controlled ordering screens. Across 34 HIAD incidents, SAGA's machine proxy was 0.760 versus 0.401 for alarm-only text; expert ratings remain pending. Evidence supports a claim-bounded prototype, not a field-qualified controller or autonomous emergency system.

Keywords: hydrogen refueling station; digital twin; external validation; HyRAM; process safety; decision support

1 Introduction

Hydrogen refuelling couples high-pressure storage, compression, cascade dispatch, precooling, flow control and vehicle-tank heat transfer. A useful digital twin must conserve mass and energy, reproduce the fast-fill transient, expose safety states and distinguish measured or simulated evidence from assumed inputs. The physical decomposition follows the principal station elements in ISO 19880-1, while a conforming light-duty fuelling controller would additionally require an authorized SAE J2601 implementation [33, 34]. Published models have addressed vehicle filling and partial or whole-station dynamics, including the

H2FillS formulation [4], optimization of cascade refuelling [5], vehicle-side pressure loss [6], and multi-zone tank heat transfer [7]. These works also show why agreement at one component boundary cannot establish the accuracy of an entire station controller. Independent field evidence is available at a coarser resolution. An NRC Canada analysis of approximately 22,593 H70T40 dispensing events at four British Columbia stations reports protocol context, normal/incomplete/fault-fill categories and plotted pressure, temperature and flow patterns [35]. The report does not release synchronized machine-readable logs, so we use it for operating-range and fault-taxonomy context rather than as a numerical full-loop holdout.

CARB tested 22 stations with an abbreviated HGV 4.3:22 matrix [36]. None passed every test; five passed fault and communication tests and four passed nine fuelling-performance metrics. Pass rates were 45.5% for general fault, 40.9% protocol fault, 50.0% communication and 18.2% fuelling performance. They motivated dispenser-specific communication faults, startup and pressure- corridor termination, T20/T30/T40 checks, and data-loss hold/resume. These training controls lack J2601 tables and HGV 4.3 thresholds. Aggregates support field relevance, not calibration.

Two real-station studies bound operating context. Genovese et al. report more than 4,500 Cal State LA HRFF events and over 8,800 kg dispensed [40]; Samuelsen et al. report fill-level pressure, temperature, mass and time summaries from the UCI NFCRC station [41]. Neither public record includes synchronized pressure, temperature and mass-flow rows with reuse terms, so neither supplies a numerical holdout.

The public PRHYDE deliverable provides a broader protocol and experiment context: four instrumented tanks (240 L H70, 350 L H50, 322 L H35 and 165 L H70) were tested at ZBT and Nikola across pressure-ramp and delivery- temperature conditions citeprhyde2023. Its public artifact contains test matrices, plots and aggregate model errors, but not the synchronized logger arrays needed for an untouched numerical holdout. An earlier JHFC/NEDO study likewise reports six practical 35/70 MPa filling conditions and station-supplied pressure/temperature measurements citemonde2012; the current public article and project archive do not expose the underlying machine-readable runs. These sources therefore support operating-range context and data-request targeting, not a new external pass claim.

Safety-oriented twins add a second model family. Release, dispersion, jet-fire, overpressure and quantitative-risk calculations are commonly delegated to a consequence package such as HyRAM+ [8, 9]. A third layer now uses large language models (LLMs) to turn process and consequence results into operator-facing explanations. The layers have different evidence requirements. Passing unit tests cannot validate thermodynamics; matching a public consequence API cannot validate site geometry; and fluent text cannot demonstrate safe or useful advice.

This work asks five separate questions:

1. How accurately does the vehicle-tank submodel reproduce public fast-fill experiments when measured boundary conditions are imposed?
2. Does the complete controller–cascade–precooler–vehicle loop meet predeclared engineering error screens?
3. Does the source-depletion and direct-aperture model meet prospectively specified screens on independent ambient hydrogen blowdown experiments?
4. Does the production adapter preserve the inputs and outputs of the exact installed HyRAM+ package?
5. Can process- and evidence-linked SAGA responses improve incident decision support

relative to deterministic alarms without increasing unsafe advice?
The first four questions have reproducible results. The fifth has a frozen protocol but no external expert result at the time of this draft.

2 Materials and methods

2.1 Digital-twin boundary

The virtual station represents tube-trailer supply, a three-stage compressor, low-, medium- and high-pressure storage banks, two independently dispatched dispenser circuits, pre-cooling, hose line-pack and Type-IV vehicle vessels. Sampled supervisory and safety logic is coupled to continuous mass and energy balances. Process operations remain event-driven: the state is held when no transfer, compression, venting or incident request is active. The visual scene, simulated sensors and emergency-action interface are research and training components; they are not an as-built survey or a certified safety instrumented system.

The process model owns pressure, temperature, inventory and mass flow. A release snapshot, including source state and explicitly selected aperture, is passed to the consequence layer. The LLM is called only after deterministic process, alarm and available consequence fields have been assembled. Sensor analysis, the main conversational interface and the separate SAGA retrieval service have isolated API contracts so that a command request is not silently replaced by a periodic status report.

2.2 Vehicle-tank model

The Type-IV surrogate uses gas, liner and composite-shell energy states. For gas mass m_g, gas internal energy U_g and effective volume V_{eff}, real-gas properties close the state through density and specific internal energy:

$$\rho_g = m_g / V_{\mathrm{eff}}, \qquad u_g = U_g / m_g, \tag{1}$$

$$(p_g, T_g, h_g) = \mathcal{P}_{\mathrm{H_2}}(\rho_g, u_g). \tag{2}$$

The property operator $\mathcal{P}_{\mathrm{H_2}}$ is implemented with CoolProp; no ideal-gas density substitution is used. The balances are

$$\frac{dm_g}{dt} = \dot{m}_{\mathrm{in}} - \dot{m}_{\mathrm{out}}, \tag{3}$$

$$\frac{dU_g}{dt} = \dot{m}_{\mathrm{in}} h_{\mathrm{in}} - \dot{m}_{\mathrm{out}} h_g - U A_{gl}(T_g - T_l), \tag{4}$$

$$C_l \frac{dT_l}{dt} = U A_{gl}(T_g - T_l) - U A_{ls}(T_l - T_s), \tag{5}$$

$$C_s \frac{dT_s}{dt} = U A_{ls}(T_l - T_s) - U A_{sa}(T_s - T_{\mathrm{amb}}). \tag{6}$$

The fixed volume is scaled from 0.122 m^3 per 4.7 kg nominal capacity because machine-readable vessel geometry was not present in the public overview. This is an explicit source of structural uncertainty.

2.3 Public fast-fill data and frozen split

Five checksum-verified archives from H2Protocol/Powertech Labs were normalized into 36 SAE J2601 Tables Method fills [3]. The model uses the experimental clock and is interpolated to measured timestamps without dynamic time warping. An experiment-only mass-closure screen compares integrated measured flow with the nominal inventory change inferred from reported state of charge (SOC). The admissible ratio was fixed at 0.8–1.2; one repeat-fill case was excluded because its ratio was 1.428. This screen contains no model prediction.

Closed-loop mass-flow timing diagnostics subsequently exposed a normalization error in development case H2P-L29. An isolated 1.526 g s^{-1} pulse at workbook time zero caused the former first/last-threshold rule to prepend 369.5 s of idle data to the sustained fill. A disclosed amendment clusters above-threshold samples using a 15 s maximum gap, discards terminal clusters below the larger of 0.005 kg or 0.2% of clustered mass, and preserves meaningful multi-segment fills and their internal gaps. The corrected H2P-L29 trace contains 399 samples over 199 s instead of 1,143 samples over 571 s. All tank results below were regenerated after this correction; the earlier normalization is superseded.

Every third laboratory test (3, 6, …, 36) forms a frozen 12-fill validation set. The remaining 23 admissible fills estimate two global parameters: effective volume multiplier and gas-to-liner heat-transfer multiplier. The fitted values were 1.05273 and 31.46066, respectively. Optimization used bounded nonlinear least squares in log-parameter space. No case-specific parameter was fitted. Case-level uncertainty was estimated with 10,000 nonparametric bootstrap resamples using seed 2601.

As an independent operating-range reference, the 3Emotion programme reports multi-year operator logs from five 350-bar bus refuelling stations and 34 buses, including aggregate fill mass, duration, average flow, availability and utilisation indicators [26]. The reported overall means (14.62 kg per fill and 10.28 min) are retained in the repository's public-data inventory as an aggregate benchmark. They are not used as a synchronized time-series holdout: the public record does not release common-time-base pressure, temperature, mass-flow and valve-state rows. This distinction prevents an operating-range comparison from being presented as full-loop physics validation.

For a variable y with n aligned samples, the principal metric is

$$\mathrm{RMSE}(y) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i^{\mathrm{model}} - y_i^{\mathrm{measured}})^2}. \tag{7}$$

Pressure, gas temperature and density-based SOC were evaluated separately.

2.4 Complete closed-loop evaluation

The full loop adds a sampled pressure-ramp controller, PI flow command, cascade selection, finite valve motion, line-pack, precooling and station-to-vehicle hydraulics. The global tank parameters were frozen before this evaluation. A development-only adaptive search selected one dispenser flow-area multiplier of 2.0 on eight declared development fills; the initial grid was extended only after its optimum fell at the upper bound. Density-based SOC used each experiment's nominal 35 or 70 MPa working pressure.

Initial comparison exposed three structural errors: flow-limited anti-windup could reverse the integral while pressure error remained positive; a fixed $-20\,^{\circ}\mathrm{C}$ precooler trip stopped warmer delivery classes; and cascade selection could regress to a lower bank.

The corrected implementation uses conditional-integration anti-windup, a delivery-class-dependent trip while preserving the default T40 limit, monotonic low-to-medium-to-high dispatch, and independent dispatch state for the two dispenser circuits. No fitted parameter was changed after these corrections.

Engineering screens were fixed at pressure RMSE $\leq$ 5 MPa, temperature RMSE $\leq$ 10 °C and absolute final SOC error $\leq$ 5 percentage points. These are project screens, not SAE conformance limits. Eleven comparison fills had already been inspected before the structural repair; their later results are therefore internal confirmation rather than an untouched holdout.

An additional Powertech Labs MC Default bench-test archive was then used as a prospective external holdout. Before any selected workbook outcome was opened, the archive SHA-256, eight eligible workbook paths, model commit, fitted tank parameters, dispenser multiplier, exclusion rules and the three engineering screens were committed. The frozen controller accepts one constant APRR, whereas the publisher supplies a pressure–time schedule. We therefore mapped each schedule to its endpoint-equivalent average pressure ramp. This disclosed approximation tests transportability of the frozen implementation; it is not an implementation or certification of the proprietary MC Formula. No case-specific fit, post-freeze tuning, dynamic time warping or failed-case exclusion was permitted.

Eight de-identified station-side traces were split 70/30 using attested medium/high storage pressure and compressor state. Eligibility required holdout restarts to exceed the fitted dwell. A clock screen found 12 pressure–flow–storage cohorts among 33 tables; none had controller state, temperature and vehicle channels required for full-loop validation.

2.5 Prospective ambient blowdown validation

The source-depletion and direct-aperture release model was independently evaluated with the PRESLHY E3.1 DisCha dataset [12]. Before any numerical Excel outcome was opened, the protocol and implementation hashes, inclusion rules, pressure interpretation, time alignment, endpoints, thresholds, bootstrap seed and negative-result policy were committed. The evaluation included all eligible 300 K workbooks from the 0.5, 1, 2 and 4 mm packages. The facility has a 2.815 L stainless-steel vessel; initial absolute pressure and internal gas temperature were taken from measured channels. Because the selected workbooks contain no ambient-pressure channel, 101.325 kPa was substituted and flagged. A terminal near-zero vessel-pressure signal was interpreted as gauge pressure. Published synchronized time, whose zero is the first significant nozzle-line pressure rise, was retained without fitted shift or time warping.

The frozen model used a rigid adiabatic 2.815 L control volume, instantaneous full opening, nominal circular aperture and pre-existing discharge coefficient 0.8. It did not model vessel-wall heat transfer, valve travel, release-line inventory, pipe backpressure or two-phase flow. Evaluation began at 0.1 s and ended at the first measured pressure at or below 1.05 times ambient or the record end. The two case screens were pressure NRMSE no greater than 10% of initial absolute pressure and time-to-50%-initial-gauge-pressure relative error no greater than 20%. At least 12 cases spanning three nozzle and three pressure groups were required; the aggregate claim required at least 70% to pass both screens. The case was the unit for 10,000 bootstrap resamples with seed 20261003. Eligible integration or property-domain failures were retained as failed cases.

The negative E3.1 result was followed by a development-only revision. It used a real-

gas equation of state, maximum isentropic mass flux, the published 28 kg stainless-steel wall, a lumped wall energy balance, Churchill–Chu internal natural convection and a 6 W m^{-2} K^{-1} external coefficient. The global discharge coefficient remained 0.8 and no case-specific fitting was allowed. Because the same 22 records informed this revision, its performance is diagnostic rather than confirmatory. The model and evaluator were then hash-locked before numerical access to the separate PRESLHY E5.1 ignited-discharge campaign [13]. The primary window required measured initial gas temperature 280–330 K, initial pressure 0.5–21 MPa, at least four experiments, two nozzle groups and two pressure groups. The original two case screens and 70% aggregate rule were retained. Complete source-pressure traces, including post-ignition samples, were evaluated; failed integrity or model evaluations were retained.

A second transfer test used the independent INERIS/CEA 25 L Type-IV campaign near 90 MPa [14]. Before numerical figure values were read, the model, evaluator, three-series minimum and endpoints were hash-locked. Figure 5 mass-flow curves were digitized independently from PDF vector paths and a 4x raster render; their mean was retained and half their difference plus half-pixel resolution defined digitization uncertainty. Table 1 supplied measured source temperature at the same pressure states. The primary window was 2–95 MPa, 220–330 K and at least eight states in each of the 1, 2 and 3 mm series. Series screens were mass-flow NRMSE no greater than 15% of measured peak and median absolute percentage error no greater than 20%; at least two of all three eligible diameter series had to pass. The pressure axis was treated as absolute; adding 101.325 kPa was predeclared as a non-controlling basis sensitivity.

A third transfer test used the independent Sandia/SRI blowdown reported by Schefer et al. [15] and the experimental Figure 3b coordinates distributed with HyRAM+ 6.1. The evaluator and endpoints were committed before any numerical coordinate was opened, although an approximate 100 s total duration was already visible in secondary descriptions and was disclosed. The locked model was a well-mixed adiabatic 0.098 m^3 vessel, initially at 15.513 MPa and 315.15 K, discharging through the published 3.175 mm controlling manifold restriction with the public HyRAM+ benchmark value $C_d = 1.0$. At least 15 points were required. All three screens had to pass: peak-normalized mass-flow NRMSE $\leq$ 15%, median absolute percentage error $\leq$ 20% above 10% of measured peak, and half-peak-time error $\leq$ 20%.

A fourth transfer test used the different 2007 Sandia/SRI pressure-decay experiment [16]. Its evaluator, code and data hash were committed before any numerical Figure 4 coordinate was opened. The locked model was a well-mixed adiabatic 1.234 m^3 vessel initially at 43.1 MPa and 290 K, discharging through the published 5.08 mm restriction with $C_d = 1.0$. At least 15 points were required. All three screens had to pass: initial-pressure- normalized RMSE $\leq$ 10%, median absolute percentage error $\leq$ 15% above 10% of initial pressure, and half-pressure-time error $\leq$ 20%.

A fifth transfer test used the scaled hydrogen release from Ekoto et al. [18]. Its evaluator, code and data hash were likewise committed before numerical Figure 3 coordinates were opened. Public apparatus inputs were 13.45 MPa, 297 K, 3.63 L, a 3.56 mm restriction and $C_d = 0.75$. The mass-flow NRMSE, median error and half-peak-time limits were the same 15%, 20% and 20% screens used for the 2006 transient, and all three were required.

Residuals motivated an apparatus-resolved candidate with source and line/manifold mass–energy volumes, finite valve travel and bidirectional inter-volume restrictions. It exposes line states and boundary flows; tests require whole-network mass and open-system

energy closure. Line-volume count declares apparatus topology rather than mesh refinement. No qualifying independent logger archive has been received, so it is excluded from validation pass counts and does not alter frozen results.

2.6 Consequence-model verification

HyRAM+ receives immutable release state and geometry from the twin. The production adapter reports (i) the directional centreline distance to 4 vol% hydrogen for an unignited jet, (ii) sampled 5 kW m^{-2} jet-fire heat flux and (iii) sampled 5 kPa unconfined overpressure. The latter two distances are bounded by the configured observation coordinates and are not regulatory setbacks.

Verification used the exact public HyRAM+ v6.1 tag and commit `b45abf9a6d99 5951311be6aad836f1874e4d420b`. Installed Python-source hashes were compared with the clone. Sandia's upstream plume, heat-flux and unconfined overpressure validation tests were executed unchanged. Three station-scale parity cases then called both the production adapter and the public HyRAM API and compared every translated field with relative tolerance 10^{-12} and absolute tolerance 10^{-9}. A separate display contract preserves the sampled thermal/blast radius, the directional 4 vol% plume length and width, and the release angle without unit conversion. Browser regression tests prevent these physically different quantities from being collapsed into one spherical extent.

2.7 Incident decision-support evaluation

The incident study population is the hydrogen-refuelling-station subset of the European Commission JRC HIAD 2.2 database [24]. Of 34 identified HRS events, 33 contain at least one emergency action, lesson or corrective field. Only a coordinator- approved historical observation is supplied to the system; recorded response and lesson fields are withheld as reviewer reference material. A deterministic pre-screen highlights exact phrase overlap and possible completed-response sentences but cannot approve a case. A non-rating coordinator must inspect all holdout vignettes; the approved input is then checked against the original case set and SHA-256 locked before response collection. For independent actual-accident provenance, 23 public KHK reports cover 26 incident codes [25]. Citation-only metadata mapped eight response families: 22 reports (25 codes) traversed canonical process, detector, safety and consequence recipes; one KOH-electrolyte release was excluded. All eight recipes passed. For vent and relief guidance, a HyDelta controlled-flare report supplied safeguards: gas pre-pressure permissive, paired-valve leakage test, low/high line-pressure trip and combustion-air control [50]. Reported conversion, NO_x and 34 vol% nitrogen results remain burner-specific; they do not validate the release model, site distance or ignition of an ordinary vent stream. As a separate provenance and integration check, controlled categories derived from the public HIAD fields were joined to staged response plans without retaining source prose. All 34 cases mapped to a registered family and eight action categories were covered. The predeclared metadata concordance matched 33; one no-release record conflicted with a release family and remained a limitation. Nine canonical recipes traversed process state, detection, safety logic and native HyRAM where applicable. Twenty-nine cases had direct physical coverage, four used a compressor proxy, and one structural-damage case remained response-only; all recipes passed. Incident prose set no parameters.

Before collecting any cohort model outputs, a separate retrospective machine bench-

mark froze the complete 34-event public HRS cohort, one integrated direct SAGA response per event, a deterministic alarm-only comparator, a five-stage response contract and eight coarse action categories. Historical emergency actions, lessons and corrective measures were withheld from the model-visible context. The primary development proxy was the mean of expected action-category recall and five-stage response coverage. A paired event-level bootstrap used 10,000 replicates and seed 20261008. This proxy tests response structure and coarse lexical agreement only; it does not test action correctness, operator usability or safety. Because the category taxonomy existed before this benchmark and the same public cohort is used for development, it is neither an independent holdout nor a substitute for expert review.

A post-outcome audit reused guarded answers without provider calls. On cases with public action categories, it measured reference precision/F1, non-reference categories and word count. Incomplete HIAD fields make these burden metrics, not correctness or safety endpoints.

The frozen protocol compares a deterministic alarm, three direct one-pass SAGA responses and three SAGA responses with standards-document retrieval. Output is masked and randomized. Three independent reviewers with documented hydrogen/process-safety experience score situation accuracy, immediate-action correctness, priority, stabilization/restart, prevention, evidence grounding and operator usability from 1 to 5. Critical omission and unsafe advice are separate binary endpoints. The event is the unit of inference. Repeats and reviewers are averaged within event before paired bootstrap confidence intervals and Wilcoxon signed-rank comparisons are calculated. Failed provider calls remain as masked failure responses for scoring rather than being removed or silently retried. Because the 24-event holdout is fixed by the eligible public-incident population, a seeded pre-outcome sensitivity simulation reports Wilcoxon power over standardized paired effects. Repeats do not increase the inferential event count, and a zero-event exact upper confidence bound is retained for unsafe-advice outcomes. No LLM-as-judge score is accepted as the primary endpoint. Holdout responses and ratings have not yet been collected. The applicable institutional ethics determination for expert-participant research must be recorded before reviewer recruitment or rating collection.

3 Results

3.1 Measured-boundary tank validation

Table 1 reports the frozen split. Validation errors were lower than calibration means for all three primary quantities, although uncertainty remains wide across heterogeneous fill conditions. Figure 1 shows aligned pressure and temperature predictions; Figure 2 exposes the case-to-case error distribution rather than only an aggregate value.

Table 1: Type-IV tank-model results. Parentheses are case-bootstrap 95% confidence intervals for the mean.

Split	Cases	Pressure RMSE (MPa)	Temperature RMSE (°C)	SOC RMSE (percentage points)
Calibration	23	4.529 (3.237–6.126)	7.889 (6.072–9.870)	4.455 (3.517–5.455)
Validation	12	3.841 (2.507–5.835)	4.833 (3.707–6.113)	3.850 (2.921–4.909)

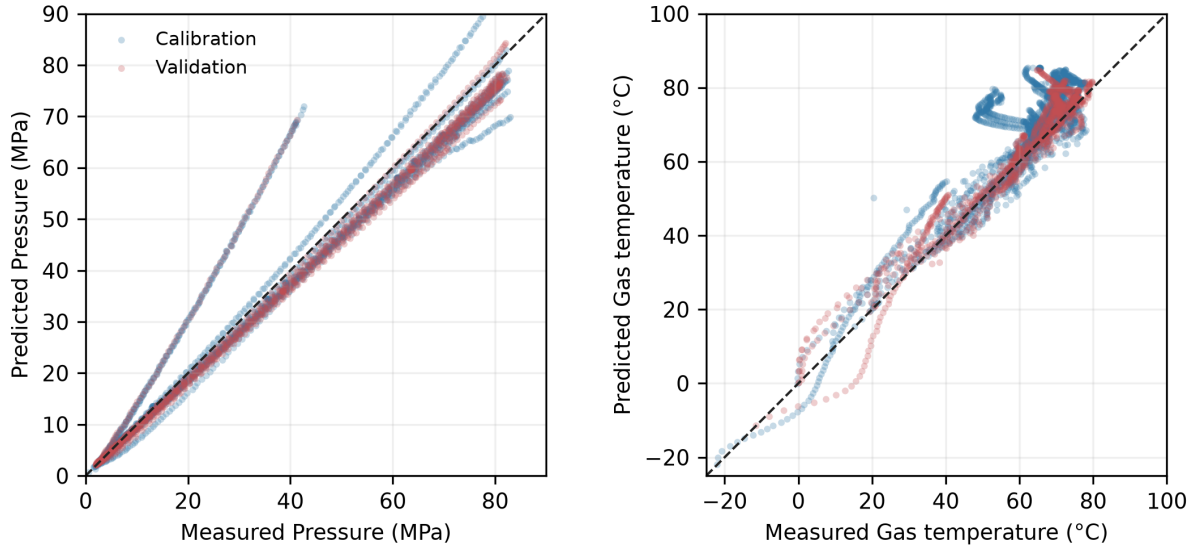


Figure 1: Measured and predicted tank pressure and gas temperature on the experimental time bases. Points are semi-transparent so repeated trajectories remain visible. The dashed line denotes equality.

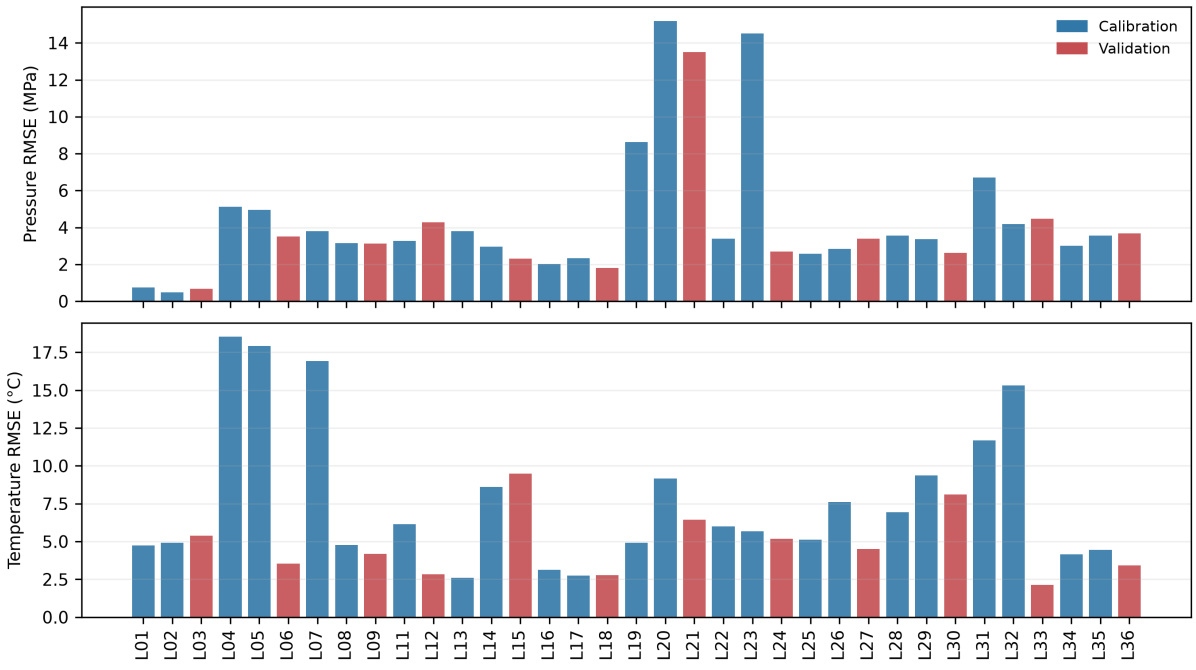


Figure 2: Case-level pressure and gas-temperature RMSE. The frozen validation cases are every third laboratory test; L10 was excluded by the pre-model mass-closure screen and therefore has no bar.

The fitted gas-to-liner multiplier is large and the optimizer's reported first- order optimality (3115.9) indicates a shallow or imperfectly resolved optimum even though its termination flag was successful. The fit should therefore be treated as an effective heat-transfer calibration, not an identified physical coefficient.

The 15-workbook CC BY 4.0 MetHyTrucks Groups A–C release provided 58,440 monotonic 0.5-s samples [44, 43, 45]. Hashes and sizes matched. Six files contained flow and cumulative mass. Nine of 18 sessions changed scale mass; 8/9 met the 0.8–1.2 screen (median ratio 1.003; median absolute difference 0.48%; Fig. 3). Nine unchanged-scale sessions were unscored. D3 identifies Groups A and C as 35 MPa direct-sampling tests with nonvehicle sinks [47]. D1 lacks units, tag mapping, controller states and calibration uncertainty [46]; outcomes are post-access evidence.

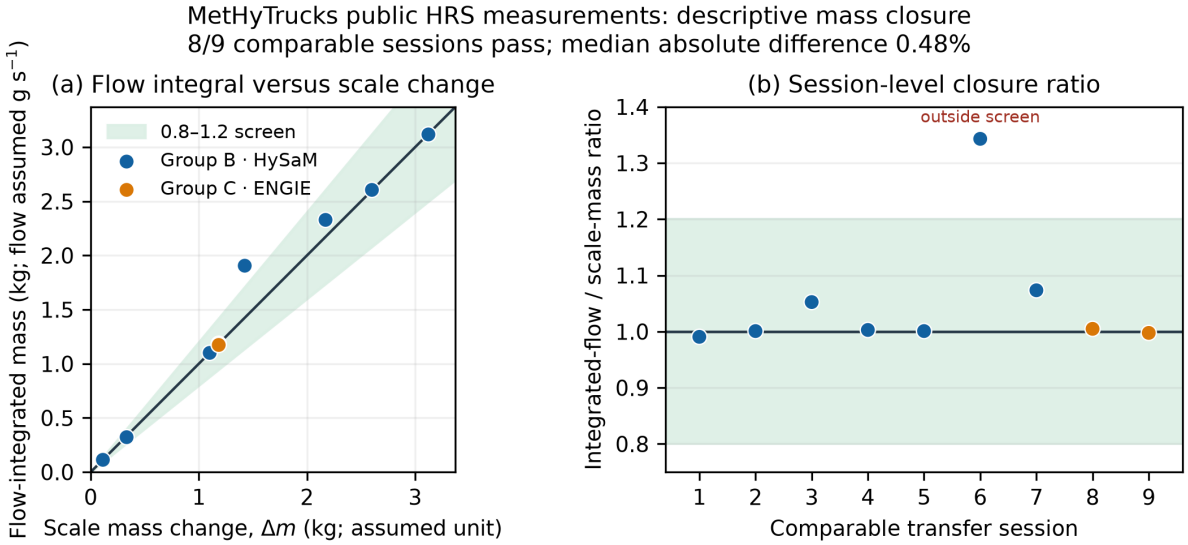


Figure 3: Descriptive flow/scale closure for nine MetHyTrucks sessions; workbook units remain audit assumptions.

Within that archive, five unfitted Hy-SaM replays had pressure and temperature RMSE of 1.287 MPa and 4.578 °C [42, 43]. All met descriptive screens, but inferred tags and missing setup mapping restrict them to a diagnostic; a 77-L alternative passed 1/5 screens. A separately frozen Group D H70 workbook contained 720 samples but lacked confirmed tank pressure, temperature, units, type and volume [48, 49]. It was retained as ineligible without replay and contributes zero full-loop cases. A frozen Type-III fill [1, 2] passed pressure (0.943 MPa RMSE) but failed temperature (19.179 K RMSE); post-outcome sensitivity cannot revise it. A post-access Woodfield Type-I replay [38] reduced filling pressure and temperature-envelope RMSE by 30.50% and 68.23%; inlet forcing stayed inactive during discharge. It is non-independent development evidence.

3.2 Negative full-loop validation result

After the preprocessing and nominal-pressure corrections, refitted tank, flow-area calibration and frozen v2 cooling sensitivity, one of eight development fills and two of eleven internal-comparison fills met all three project screens. Their mean errors improved to 6.602 MPa, 11.428 °C and 7.297 SOC percentage points for development, and 4.530 MPa,

7.819 °C and 4.625 percentage points for internal comparison. The low joint pass fractions still constitute a negative full-loop result.

The prospectively frozen MC Default holdout had already been evaluated with its pre-correction frozen model and passed zero of eight screens (Table 2). Mean pressure, temperature and SOC RMSE were 15.862 MPa, 13.230 °C and 18.082 percentage points, respectively. Five runs terminated on the model's temperature safety logic; final SOC error was negative in all eight cases and ranged from −8.42 to −50.79 percentage points. These outcomes were not rerun or used to tune the corrected model, so they remain historical external failure evidence rather than validation of the final configuration.

Table 2: Closed-loop performance. The development and already-inspected comparison sets use the corrected v2 pipeline; the MC Default set reports its prospectively frozen historical model and was not reused after correction.

Set	Pass/all	Pressure RMSE (MPa)	Temperature RMSE (°C)	SOC RMSE (percentage points)
Corrected development	1/8	6.602	11.428	7.297
Corrected internal comparison	2/11	4.530	7.819	4.625
Prospective MC Default, frozen earlier model	0/8	15.862	13.230	18.082

The measured-boundary tank result therefore cannot be used as validation of the controller, cascade schedule, precooler or dispenser boundary. The independent holdout demonstrates that the failure is not confined to the previously inspected comparison set. Systematic underdelivery and false temperature stops identify the constant-ramp protocol representation, source hydraulics and precooling boundary as redesign targets. The bench files observe vehicle and source pressures, flow, inlet temperature and SOC, but do not expose every internal three-bank valve command; they therefore cannot by themselves validate all cascade dispatch states.

An outcome-independent source-boundary diagnostic also limits interpretation. Across eight traces, third-source pressure fell 1.36–4.91 MPa, versus 30.7–58.6 MPa predicted for an isothermal 0.35-m^3 bank; observed pressure and mass changes implied 2.74–8.70 m^3. Because the files omit topology and valve states, the result remains diagnostic. Prospective evaluation must supply boundary pressure and temperature or disclose the topology and valve states.

Correctly pairing co-located `Pinlet` and `Tinlet_G` reduced the post-outcome mean temperature RMSE from 25.884 to 9.052 °C; pressure and SOC RMSE were 4.376 MPa and 5.583 percentage points, with zero of eight joint passes. Two non-fitted geometry rules reached four of eight passes but were not selected. Prospective tests now require exact vessel geometry and temperature-sensor position.

A superseded development-only sensitivity first scaled hydrogen–coolant and chiller conductance by factors 1, 2, 4 and 8 and selected its upper bound. After the parser, tank-fit, flow and nominal-pressure corrections, a replacement protocol was frozen before evaluating factors 1, 2, 4, 8, 12 and 16. It again selected the upper bound, but the normalized objective decreased only from 11.051 at 8 to 10.969 at 16. The retained factor 16 produced the corrected results in Table 2; larger values were not searched. The upper-bound optimum, diminishing improvement and remaining case-level failures indicate that cooling duty alone does not explain the closed-loop error. Neither sensitivity was applied to the consumed external holdout.

To separate the protection logic from the underlying boundary-model error, a post-outcome sensitivity reran the same eight traces with both the controller and safety-PLC

vehicle-gas temperature trip raised from 85 to 95 °C. This counterfactual produced no screening passes; mean pressure, gas-temperature and final-SOC RMSE were 10.217 MPa, 14.301 °C and 12.605 percentage points, respectively, and none of the cases stopped on the temperature limit. The diagnostic is therefore evidence that the negative holdout is not explained solely by premature temperature shutdown. It is explicitly post-outcome, does not change the frozen decision, and cannot substitute for a new prospectively frozen holdout with an independently justified safety limit. The complete run metadata are archived in `research/closed_loop_temperature_stop_diagnostic_2026_10_04.json`.

3.3 Negative confidential station recharge holdout

Calibration (457,405 samples; 89 completed restarts) fitted a 290.0-s dwell. The disjoint holdout (196,037 samples; 32 restarts) contained a 176.0-s minimum, so the gate failed. The prior single-trace 265.2-s result was superseded and runtime use locked. This station-side diagnostic does not validate compressor capacity, vehicle filling or the station-to-vehicle loop.

The same attested high-stage state/pressure pairing was then re-analysed with the controller quantities separated. Across all eight records, 124 completed stop-to-restart cycles had a median pressure drop of 4.555 MPa (10th percentile 4.36 MPa). The simulator therefore uses a rounded 4.5 MPa high-bank restart band as an evidence-informed development default while leaving the failed time dwell disabled. This analysis was performed after outcome access and is not an independent holdout, safety limit, or full-loop validation result.

3.4 Prospective confined pressure-peaking screen

To test a separate consequence boundary without presenting it as station-loop validation, a second prospective protocol was frozen before accessing the raw outcomes of the HyTunnel-CS unignited pressure-peaking release archive [31]. The source contains ten synchronized pressure and mass-flow experiments (cases 2–11) in a 14.9 m^3 vented enclosure. The fixed model used source-documented vent areas and initial temperatures, the measured mass flow as input, a constant-temperature ideal-gas balance and the fixed publication value $C_d = 0.7$; no case-specific fitting or outcome-based case selection was permitted.

All ten cases were eligible. Seven passed both predeclared primary screens (peak overpressure absolute error ≤ 5 kPa and peak-time relative error $\leq 20\%$), giving 70%. This was below the frozen 80% confirmatory threshold, so the result is exploratory evidence for a confined unignited pressure-peaking submodel only. It does not validate outdoor dispersion, ignition, emergency response or the complete HRS fueling loop.

3.5 Negative ambient blowdown validation result

All 22 PRESLHY workbooks were eligible, spanning four nozzle diameters and low, medium and high initial-pressure groups. Eleven cases passed both primary screens, giving a joint pass fraction of 50.0% (case-bootstrap 95% CI 31.8–72.7%), below the predeclared 70% criterion. The direct-aperture source-depletion claim is therefore not supported for this model revision.

Six eligible cases, all in the high-pressure group and initiated at approximately 150 or 200 bar, encountered a retained thermophysical-domain error when the calculated

restriction choking state left the single-phase hydrogen table. They were counted as failures. Among the 16 numerically completed cases, 11 passed both screens (68.8%); this post hoc descriptive fraction does not replace the primary decision. Joint pass counts were 2/9 at high pressure, 6/7 at medium pressure and 3/6 at low pressure. By aperture they were 2/4, 0/4, 4/7 and 5/7 for 0.5, 1, 2 and 4 mm, respectively. These strata indicate both a runtime-domain limitation and model-form error rather than one isolated failed workbook.

3.6 Development revision and negative E5.1 holdout

Adding documented vessel-wall thermal inertia and real-gas discharge physics removed calculation failures on the consumed E3.1 set. Twenty of 22 development cases passed both original screens (90.9%; case-bootstrap 95% CI 77.3–100%). This improvement cannot validate the revision because the same experiments had already been examined and used to motivate the model change.

The frozen E5.1 holdout did not confirm the revised claim. Only three of seven campaign experiments met the primary measured-temperature definition, below the minimum four. Two of those three passed both screens (66.7%; bootstrap 95% CI 0–100%), also below the 70% decision threshold. The two successful cases were 4 mm releases near 50 bar absolute; a 2 mm low-pressure case passed the pressure NRMSE screen (9.04%) but missed the half-pressure timing screen by 305.7%. Two publisher-labelled ambient winter runs had measured pre-release temperatures of 277.28 and 273.71 K and therefore remained outside the frozen primary window. The latter and both cryogenic transfer runs triggered the predeclared significant-pressure-increase integrity check and were retained as failed evaluations. The publisher explanation labels the 2 mm run as 50 bar, whereas the workbook metadata and pre-release trace are approximately 5 bar gauge; the analysis retains the measured state and separately records the nominal label. The eligibility minimum and performance threshold both failed, so the revised source-depletion claim is not supported by this holdout.

3.7 Negative independent 90 MPa mass-flow transfer

All three INERIS/CEA diameter series met their eligibility minima, with 13, 10 and 12 paired states for 1, 2 and 3 mm, respectively. None passed both screens. Peak-normalized NRMSE was 16.27%, 16.01% and 32.62%, while median absolute percentage error was 25.07%, 15.85% and 32.88%. The 1 mm series was generally underpredicted, whereas overprediction increased for the 2 and 3 mm series. Adding 101.325 kPa to the plotted pressure did not change any decision, and no prediction lay inside the narrow digitization interval. Post-outcome descriptive least-squares effective discharge coefficients were 1.141, 0.657 and 0.533; they are diagnostics and were not used to rerun or tune the primary result. The 0/3 joint pass count rejects the independently validated high-pressure aperture-flow claim for the fixed model.

3.8 Negative Sandia/SRI transient mass-flow transfer

The Figure 3b digitization supplied 25 unique experimental coordinates. The locked model passed the NRMSE screen at 5.83% of measured peak, but median absolute percentage error was 22.85% and half-peak-time error was 28.29%. Measured and predicted peak flows

were 68.38 and 60.67 g s^{-1}; measured and predicted half-peak times were 13.35 and 9.58 s. Because all three screens were required, the external transient claim is not supported. The lower predicted peak followed by faster decay is consistent with omitted vessel-wall heat input and downstream-tube inventory, but that interpretation is diagnostic and no post-hoc parameter change enters the primary result.

3.9 Negative Sandia/SRI pressure-decay transfer

The separate Figure 4 pressure trace supplied 222 unique coordinates. The locked model reproduced the initial pressure within 0.04% and passed the half-pressure-time screen: measured and predicted times were 26.79 and 28.53 s, for 6.51% error. It failed the two curve-wide screens. Initial-pressure- normalized RMSE was 11.58% and median absolute percentage error was 27.46%. The experimental trace lost pressure much faster during the first seconds, whereas the model later fell below the measured pressure. Because all screens were required, the external pressure-decay claim is not supported.

A planned 50 L, 20 MPa penetration-release transfer using Jankuj et al. [17] was invalidated when the audit found that its numerical workbook had already been inventoried before protocol freeze. The retained post-access diagnostic also failed all three holdout screens without refitting. It is reported for transparency and supplies no additional validation claim.

3.10 Positive bounded Ekoto mass-flow transfer

The Figure 3 digitization supplied 39 unique experimental coordinates. All three prospectively frozen screens passed: peak-normalized NRMSE was 3.84%, median absolute percentage error was 12.79%, and half-peak-time error was 6.07%. Measured and predicted peak flows were 0.04483 and 0.05124 kg s^{-1}; measured and predicted half-peak times were 0.506 and 0.537 s. No post-freeze parameter or code change was made. This supports a bounded source-depletion claim for the 13.45 MPa, 3.63 L scaled release, not the enclosing warehouse phenomena or a full refuelling station.

3.11 HyRAM+ source, tests and adapter parity

The installed and upstream v6.1 Python sources matched over 38 files with aggregate SHA-256 `d5865e1ae7b5c77b985e06b45d7a5607543703c1834fc50f7c5d96f32a0c22de`. All 47 selected upstream validation tests passed, comprising 803 subtests and seven expected physics warnings. Production-adapter outputs matched independent public- API calls for all three parity cases (Table 3).

Table 3: Station-scale HyRAM+ 6.1 adapter parity. The plume distance is directional.

Case	Modelled flow (kg s^{-1})	4 vol% distance (m)	Visible flame (m)
Low bank, 1 mm	0.016613	4.339	2.326
Mid bank, 2 mm, cold	0.108305	11.382	6.057
High bank, 3 mm, override	0.296366	17.224	9.246

This verifies package translation and output indexing, but neither repeats the underlying experiments nor validates station layout or exclusion zones. A public 22-release

open-channel dataset [22] was not force-fitted because its downward 4.6 mm confined jet differs from the outdoor free-jet adapter.

The 29-channel records replayed the fixed 1.0/2.0 vol% alarm/trip and 0.5 s persistence logic; all 22 tests triggered both levels, yielding 638 sensor-case observations. In a post-access stratification, ceiling probes achieved 100% alarm/trip coverage and the highest group mean in 20/22 tests; the three floor probes nearest the jet responded at median 10.68 s versus 18.01 s at the ceiling. This supports layered ceiling plus jet-path coverage in this geometry, not independent station placement, ESD, outdoor transfer or runtime validation.

A physical-H_2 benchmark was then run under a protocol committed before the public CC BY 4.0 workbook was opened [23]. Across 45 observations, the portable sniffer and mass-spectrometer vacuum/sniffer modes achieved Spearman $\rho = 0.943/0.969/0.949$, pairwise concordance 1.000 and complete mapping. All frozen rank screens passed. Median response/reference ratios nevertheless ranged from 0.892 to 0.416, 1.086 to 0.862 and 0.058 to 0.048, respectively. Thus only monotonic ordering is supported; no calibration, placement, threshold, ESD or runtime claim is made.

On a six-sensor Sandia/SRI actual-H_2 summary [18, 19], the frozen candidate raised Spearman ρ from 0.600 to 0.943 with top-three recall 1.0. The post-access, combined-test diagnostic is not independent validation and runtime remains disabled. HyDelta was unscored because required coordinates, orientation and per-sensor numbers were absent; figures were not digitised.

3.12 Decision-support study status

The deterministic split produces a 24-case holdout casebook and nine development cases, but all vignettes remain unapproved until coordinator leakage review and qualified expert sign-off; the institutional ethics determination is also pending. The pre-outcome sensitivity simulation estimated 77.7% power at a standardized paired effect of 0.6 and 95.7% at 0.8. Even with zero unsafe-advice events among 24 events, the exact two-sided 95% upper event-rate bound would remain 14.2%. Consequently, no expert-rated SAGA effectiveness, safety, omission-rate or inter-rater-agreement result is reported in this draft. This empty primary result is deliberate: substituting model self-assessment would violate the frozen protocol.

The 34-case retrospective machine benchmark gave SAGA versus alarm-only recall of 0.524/0.189, five-stage coverage of 0.982/0.600 and proxy scores of 0.760/0.401 (paired difference 0.360; bootstrap 95% CI 0.309–0.411). A post- outcome numeric guard reduced flags from 3/34 to 0/34 and passed a 5/5 regression. Across 33 events, F1 was 0.564 versus 0.281, with longer SAGA responses. These software checks do not establish expert-judged effectiveness.

4 Discussion

4.1 What is externally supported

The strongest independent result concerns vehicle-tank thermodynamics under measured inlet mass flow and temperature. It supports use of the tank submodel for research comparisons within the tested fast-fill envelope, subject to the effective geometry and

heat-transfer assumptions. The result is strengthened by a deterministic split, experiment-only exclusion rule, global rather than case- specific fitting, trace alignment on the original clock and case-level bootstrap intervals.

The complete-loop result is informative precisely because it is negative. A visually plausible pressure ramp and a stable numerical integration are insufficient evidence for a station controller. The prospective MC Default result also shows that freezing the analysis alone cannot rescue a structurally inadequate protocol representation. The retained failures identify the missing information needed for the next experiment: dispenser valve command and position, upstream and downstream pressure at the PCV, delivered-gas temperature, bank-selection state and high-rate vehicle inlet mass flow. The eight MC Default and eleven previously inspected comparison cases are now consumed evaluation evidence and cannot be used as a new holdout. Model redesign must use development evidence or physical invariants, followed by a separately acquired and pre-frozen external set.

The negative PRESLHY result places an additional boundary on the release path. The published DisCha vessel weighs about 28 kg, whereas the frozen validation model deliberately imposed an adiabatic gas control volume with no wall heat transfer. Deep calculated cooling ultimately left the available restriction-state domain in six high-pressure cases, and several lower-pressure cases completed but missed the half-time or pressure screen. A revised model should represent the documented vessel thermal mass and physically justified gas–wall transfer, extend the thermophysical domain only where single-phase assumptions remain valid, and test valve/line dynamics as a separate effect. The 22 cases now constitute consumed development evidence. Any improved version requires a new hash-locked external holdout rather than rerating these cases as independent validation.

The E5.1 campaign supplied that prospective test, but it did not rescue the claim. Its negative decision is driven jointly by the frozen eligibility rule and performance, not by selective case deletion. The two nominally ambient winter runs illustrate why measured-state definitions must be fixed prospectively. Although the remaining two 50 bar, 4 mm cases transferred well, the primary set was too small and lacked a successful high-pressure case. E5.1 also shares the DisCha facility and investigators with E3.1, so even a passing result would have been separate-campaign evidence rather than independent-laboratory replication.

The independent French campaign then exposed a different limitation. A constant coefficient multiplying nominal aperture area cannot represent the observed diameter scaling across a 10 m delivery path: the fitted diagnostic coefficient decreased strongly as diameter increased, while exceeding unity for 1 mm. This does not justify three new coefficients. It requires an explicit valve–pipe– orifice network, geometric uncertainty and a new untouched facility for testing. The Proust curves are now consumed development evidence. Their plot-digitization uncertainty was much smaller than the discrepancies and did not determine the negative result.

A post-outcome series-conductance fit brought the consumed Proust curves inside descriptive screens, but NRMSE rose to 29.25% and 41.81% on the already-visible 3 and 4 mm Imamura series; direct-aperture $C_d = 1.0$ gave 3.79–7.60% across 1–4 mm. These development-only results indicate apparatus-specific upstream resistance and leave the negative Proust decision unchanged.

The apparatus-resolved candidate implements these missing dynamics, but remains development-only. Its protocol requires synchronized state, flow, valve position and as-built geometry from new campaign data; the consumed Proust, Schefer and PRESLHY

outcomes cannot be recycled as its holdout.

The separate Sandia/SRI transient had low aggregate NRMSE but the adiabatic model depleted flow too quickly. It motivates non-adiabatic cylinder walls and explicit 7.6 m tube line-pack; the consumed approximate-duration trace is not fully blinded.

The 2007 pressure trace makes the omitted dynamics more specific. A single adiabatic vessel state could match the initial value and half-pressure time while still missing both the rapid initial manifold transient and the heat-supported late tail. This experiment is now consumed evidence. It motivates explicit valve/manifold line-pack and wall thermal inertia, followed by a new facility- level holdout rather than post-hoc adjustment of the failed result.

The Ekoto holdout shows that the unchanged model is useful within a narrower regime: a short, small-inventory 13.45 MPa release with independently specified C_d. Its joint pass does not resolve the failed 15.5 MPa long-tail mass-flow, 43.1 MPa pressure-decay or 90 MPa diameter-scaling tests. Reporting the positive and negative transfers together defines the model's present applicability more credibly than a single validation curve.

4.2 Consequence calculations and risk communication

The consequence audit separates source identity, adapter parity and release-boundary semantics. Public dilution, heat-flux and overpressure data [20, 21, 8] support a bounded free-jet display. A post-access seven-method HRS QRA dataset [51] retained 211 rows in 46 groups and differed by up to 3.31-fold; the runtime fell inside two of four thermal and no comparable overpressure envelopes. Its 1.327 kg s^{-1} free-orifice flow exceeded the benchmark's 0.060 kg s^{-1} limit. The software records this flow-boundary approximation without tuning or setback claims.

Two literature comparisons were added. The 78-experiment Cirrone correlation reproduced the 21.9 kPa example and 24 rounded radial distances within 0.044 m [10]. The Molkov flame-length relation, supported by 123 reported experiments, gave 14.32% mean absolute percentage error over 25 transcribed high-pressure rows; 22 were within 30% [11]. Values are domain flagged. The former is cloud-centred radial overpressure and the latter visible free-jet flame length; neither is a site safety distance.

Two prospectively controlled public consequence archives were evaluated. The first contained eight real-scale 350/700 bar releases through 0.5/1.0 mm TPRD outlets in a ventilated semi-confined space [27]. Its peak mass flows were 4.04–13.00 g s^{-1} and four heat-flux channels spanned 0–57.29 kW m^{-2}. Missing experiment-to-nozzle-angle mapping precluded a model score, so these data remain descriptive thermal-boundary evidence.

The second contained 31 immediately ignited releases in a 14.9 m^3 vented enclosure [28]. After four previously opened cases were fixed as development-only, cases 2–28 formed the untouched holdout. A zero-dimensional species and energy balance used the reported common discharge and heat-transfer coefficients with measured mass-flow histories and no case fitting. The frozen rule required at least 20 cases and at least 80% with peak-overpressure error $\leq$ 2 kPa. All 27 cases passed; peak-pressure MAE was 0.672 kPa and maximum absolute error was 1.796 kPa. Secondary peak-time and trace-NRMSE screens passed 21/27 and 26/27 cases. This supports the pressure-peak endpoint for an immediately ignited vented enclosure within the tested geometry and flow range, not outdoor jet radiation, station-loop control or a site-specific evacuation distance.

A third campaign supplied 51 large-scale open-atmosphere actual-H_2 tests [29]. A protocol fixed after aggregate-only access but before case-level comparison retained 44 con-

trolled delayed-ignition cases in six instrumentation-era/nozzle/ignition strata. Pooled Spearman correlation passed at $\rho = 0.572$, whereas stratified correlation (-0.023), pairwise concordance (0.421) and top-third recall (0.294) failed frozen 0.45/0.65/0.50 thresholds. Predictions occupied 22.091–22.740 kPa versus measured maxima of 0.220–235.390 kPa. Source pressure and nozzle diameter therefore did not preserve geometry-controlled severity ordering; no parameter or threshold changed. Raw-file headers confirmed four kPa channels at 500 kHz. The archive omits case-resolved obstacle, ignition and sensor coordinates and ambient records; a frozen input contract prevents fitting them to consumed peaks.

4.3 Evidence-gated language-model assistance

The LLM is positioned after deterministic calculations and receives structured sensor provenance, scenario status, consequence fields and uncertainty flags. This order prevents text generation from fabricating a missing damage radius. The planned three-arm incident study will evaluate blinded safety ratings, unsafe advice and critical omissions.

All 34 HIAD metadata cases route to registered plans. Canonical runtime replay adds process-to-detector traceability for 29 direct and four partial-proxy cases; structural damage remains response-only. SAGA produced more requested stages and action-category terms than alarm-only text, and the API numeric guard blocked unsupported value–unit phrases. These are software-traceability results; no independent reviewer assessed unsafe omissions.

4.4 Limitations

All sensors and actions are simulated. Station geometry, weather, occupancy, valve coefficients, compressor maps and proprietary J2601 control remain unvalidated assumptions. Most release-transfer gates failed; the positive Ekoto result covers only a small, short-duration source. ELVHYS provides cryogenic pressure-peaking observations, but its post-access replay is provenance/timebase evidence because vehicle and controller telemetry are absent [32]. HIAD reports are retrospective, and the 34-case machine proxy and post-outcome guard rerun cannot replace the pending expert study.

5 Conclusions

A hydrogen-station safety digital twin was evaluated with separate evidence gates for component physics, full-loop behavior, consequence-software translation and LLM decision support. The measured-boundary Type-IV tank model achieved validation mean RMSE values of 3.841 MPa, 4.833 °C and 3.850 SOC percentage points. The complete 15-workbook MetHyTrucks intake added 58,440 synchronized public experimental samples and eight descriptive flow–mass closure passes, while its documented sampling-only boundary, metadata gaps and prior outcome access precluded a prospective full-loop claim. The complete loop and prospective ambient-blowdown holdouts did not meet their declared screens. An eight-trace confidential station dwell rule also failed its holdout and remains disabled. Independent release tests exposed pressure-, diameter- and apparatus-dependent errors that an aperture-only source cannot resolve. The positive Ekoto result passed all three frozen mass-flow screens but is bounded to its 13.45 MPa, 3.63 L source.

The HyRAM+ adapter reproduced its installed v6.1 API contract, and all 27 ignited-enclosure holdouts met the 2 kPa peak-error screen (MAE 0.672 kPa), supporting only those component endpoints. The retrospective 34-event benchmark favored direct SAGA over alarm-only text on the frozen structural proxy (0.760 versus 0.401). A post-outcome direct-API guard reduced lexical flags from 3/34 to 0/34; a 5/5 regression then preserved equivalent grounded units while rejecting an absent value. Expert study remains pending. The evidence supports a reproducible research prototype and validation workflow, without supporting SAE-compliant fuelling control, field safety certification or autonomous emergency management.

Data and code availability

Source code, study scripts, frozen machine-readable evidence and documentation are available under the MIT License at `https://github.com/lyullee/hydrogen-station-sim`; archived release DOI: 10.5281/zenodo.23084421. Third-party archives retain their original terms; HyRAM+ remains GPL-3.0.

Confidential station records are not deposited. The manuscript reports only custodian-approved de-identified aggregates and uncertainty intervals; a secure editor/reviewer inspection route is provided when permitted. A measurement schema, controlled-access protocol and synthetic traces are supplied for public reproduction. If raw inspection cannot be arranged, claims are limited to the published aggregates and publicly reproducible component evidence. The public manuscript intentionally withholds the operator, site, country, calendar dates, manufacturer, equipment model numbers, tag names and detailed geometry; those fields are retained only in the owner-controlled record.

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